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Final Report

Attendance Tracking management system

In collaboration with: Orange Digital Center

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Table of Contents

1- Project Overview	3
1.1- Introduction.....	3
1.2- Project Description	3
1.3- Problem Definition.....	3
1.4- Vulnerabilities and Existing Solutions	3
1.5- Proposed Solution	4
1.6- Conclusion	4
2- Product Specifications	5
2.1- Introduction.....	5
2.2- Project Management.....	5
2.3- Product Backlog.....	7
2.4- Sprint Backlog.....	9
2.5- Requirements Specification.....	10
2.6- Project Considerations	12
2.7- Conclusion	14
3- Product Design.....	14
3.1- Introduction.....	14
3.2- High-Level Design	15
3.2-1. Architecture Overview.....	15
3.2-2. Component Diagrams	16
3.3- Detailed Design.....	116
3.3-1. Sequence Diagrams	19
3.4- UI Design	24
3.5- Conclusion	24
4- Development and Prototype Solution	25
4.1- Introduction	25
4.2- Development Details.....	25
4.3- Development Support.....	26
4.4- Conclusion.....	26
5- Tests, Results, and discussions	28
5.1 Introduction.....	28
5.2 Test Cases	Error! Bookmark not defined.
5.3 Discussion	28
5.4 Conclusion	30
6- General Conclusion & Future work.....	31

List of Figures

Fig. 1- Standard 4” connection- Contact pressure along the connection- Zoomed...Error! Bookmark not defined.

List of Tables

Tab. 1- Comparing results using the new material test data. Error! Bookmark not defined.

1- Project Overview

1.1- Introduction

This report outlines the implementation of an attendance tracking system. It details the steps taken from the initial specification of requirements to the final implementation, including the methodologies employed and the work plan for the term. The report concludes by discussing the limitations of the system and any challenges encountered during the process.

1.2- Project Description

//desc ODC Reef.!

Our project is a web-based platform with two types of users; The admin that works in orange development center and learners that will check in using the attendance tracking management system. The system uses facial recognition technology to accurately identify learners and record their attendance in real-time. The Attendance Tracking Management System with Facial Recognition would benefit the organization by reducing the time and effort required to manually track attendance, reducing errors and inconsistencies, and providing valuable insights into attendance patterns. Facial recognition technology would also eliminate the need for manual check-ins, increasing efficiency and accuracy.

1.3- Problem Definition

The need for this platform arises from the difficulties faced by orange development center to manage the attendance for 30-40 learners as well as the tardiness and early exits of the learners since there is no fixed time to take the attendance. In fact, they are facing many issues to keep track of the attendance every morning since they do not have access to the data at the ODC level. The current methods of taking attendance are prone to errors and are time-consuming. This leads to the need for a system that can quickly and accurately track attendance using facial recognition technology.

1.4- Vulnerabilities and Existing Solutions

The existing solution is not optimized as the orange development center must manually track the attendance of the learners which is a very slow and time-consuming process. Moreover, the learners check in using a badging system that does not handle the data at the ODC level, which is not very efficient. This lack of efficiency leads to various problems in the organization.

1.5- Proposed Solution

The solution that Orange development center proposed to implement is an attendance tracking system using facial recognition. A facial recognition solution that can match learner's face to their pre-uploaded image and determine the time and location in which it was taken. A cohort management solution that allows the admin user to create and manage different cohorts while being able to set a location, start and end dates for each cohort with a user-friendly interface that allows learners to check-in at the arrival to ODC while ensuring that the system is secure and scalable.

web server:

- Facial recognition.
- Metadata extraction from image to determine time and location where the picture was taken.
- Detect who the learner is and mark him present if the location detected from the picture is at the ODC.

Admin Dashboard:

- Cohort management module where the admin can add new cohorts, set a start/end date and location for the cohort.
- Learners onboarding module to assign learners to different cohorts.
- Upload images on day one of the training for each learner, to be used for facial recognition.
- Detailed attendance reports every day of the training (present/absent + check-in time).

Client Interface:

- A simple interface where the learner can take a picture and submit it upon arrival to the ODC. can be an app to install in one of our terminals or a web interface that every learner can access on the Client WebApp.

1.6- Conclusion

In this portion of the report, the project was presented in its entirety and its goals were outlined. Additionally, our solution was compared to existing ones.

2- Product Specifications

2.1-Introduction

In this section, we will present the details of the ODC Facial Recognition Attendance Tracker's product specifications, which include the project management strategy, product and sprint backlogs, and requirements specification. The purpose of this section is to establish a well-defined and straightforward plan for the efficient creation and implementation of the app. We will also cover important project considerations and limitations that need to be considered. By laying out this information, our goal is to offer a complete understanding of the Facial Recognition Attendance Tracker app's requirements and objectives.

2.2- Project Management

The development of software using the Agile methodology is a process that involves iterations, prioritizing customer needs, and emphasizes teamwork, flexibility, and continuous improvement. For this project, we will be using the Agile methodology to guide the software development process. This means that we will work in short sprints, regularly gathering feedback from stakeholders and end-users, and adjusting our approach as necessary. We will have frequent meetings and maintain open communication to ensure that all members of our team are working toward the same goal. By following the Agile principles, we will be able to quickly adapt to changing requirements and deliver high-quality software that meets the evolving needs of our users. Our objective is to create a software solution that is efficient and effective by sticking to the Agile principles. Ref 2

2.2-1. Methodology

To oversee the development process, we used the Scrum agile methodology. Scrum is a popular paradigm for software development projects given that it places a strong emphasis on teamwork, adaptability, and continual improvement. In this part, we give a brief overview of the Scrum ceremonies we used and how the framework contributed in the management of the project. Our development cycle was broken up into a number of 2-week sprints, with each sprint signifying a new development stage. The team met often during each sprint to discuss progress and map out the following stages. At the conclusion of each sprint, we had a sprint review to assess our achievements and pinpoint any areas where we might have done better.

In summary, the Scrum framework had a significant role in the overall success of our project, and we are optimistic that it will continue to be a useful tool for projects in the future.

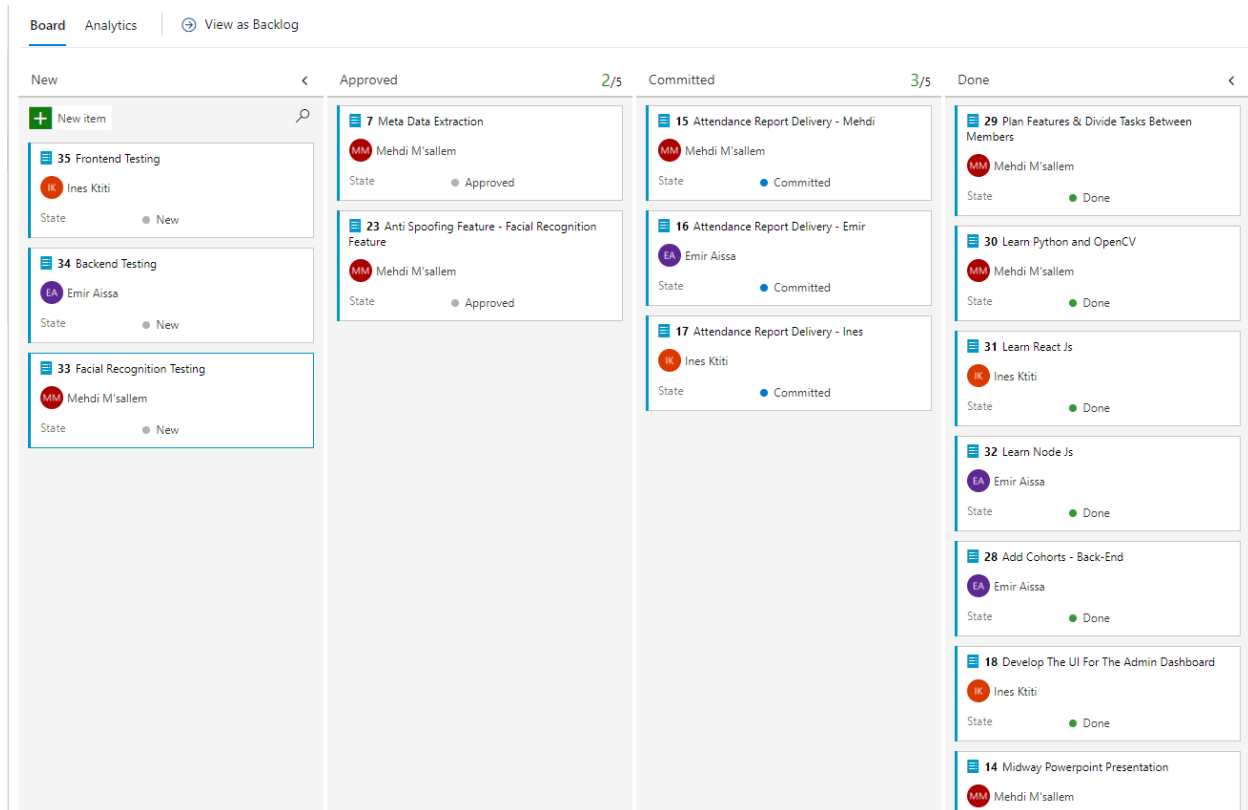
Scrum Roles:*Product Owner: Heni Mezrani**Scrum Master: Mehdi M'sallem**Development Team:**Ines Ktiti**Emir Aissa**Mehdi M'sallem***2.2-2. Timeline**

We have divided the project into features, where each feature has a set of user stories assigned to it. We did this to have a logical representation of the project and to define its priorities. We split the development period into X 2-weeks sprints. In this subsection, we present how we conducted the Scrum ceremonies.

Sprints	From	To
Sprint 1	20/02/2023	06/03/2023
Sprint 2	07/03/2023	21/03/2023
Sprint 3	22/03/2023	05/04/2023
Sprint 4	06/04/2023	19/04/2023
Sprint 5	20/04/2023	05/05/2023

Table 1 – Sprint Table

2.3-Product Backlog



User Stories	Priority
As an admin, I should have access to my own dashboard	M
As a user, I should be able to upload my picture so that that I can mark my presence	M
As an admin, I should be able to remove a learner from the cohort	M
As an admin, I should be able to set start/end times for the cohort	M
As an admin, I should be able to create a new cohort	M
As a user, I should be able to access my attendance report	M
As an admin, I should be able to have access to an analytics dashboard	S
As an admin, I should be able to access a daily report so that I can check attendance	S
As an admin, I should be able to add a learner to the cohort	M
As an admin, I should be able to upload the image of users	M
As a system, I can Utilize facial recognition technology to match the learner's face to their pre-uploaded image.	M
As a system, I can Use metadata extraction to determine the time and location of the image taken.	M
As a system, I can Verify that the location detected from the image is at the ODC and mark the learner as present.	M

Table 2 –Complexity Estimation Table

2.4-Sprint Backlog

Feature 1: Facial Recognition

User Story 1: As a Learner, I want to be able to use the following features:

Task/Assignee	Emir	Mehdi	Ines
Upload the Learner's photo to the database		X	
Facial Recognition		X	
Invokes Camera & Biometrics		X	
Scan the user's face		X	
Log the learner in		X	
Mark Attendance		X	

Table 3 – Sprint Backlog Table Feature 1

Feature 2: Admin Dashboard

User Story 1: As an Admin I want to be able to add and edit the following features:

Task/Assignee	Emir	Mehdi	Ines
Add student to cohort	X		X
Add cohort to list	X		X
Delete cohort	X		X
Delete student	X		X
Update cohort	X		X
Update student	X		X

View cohort list	X		X
View students from a specific cohort	X		X
Admin-panel back end	X		

Table 4 – Sprint Backlog Table Feature 2

2.5-Requirements Specification

2.5-1. Functional Requirements

- The system must be able to capture an image of the learner's face upon check-in.
- The system must be able to compare the captured image to the pre-uploaded image of the learner.
- The system must be able to determine the time and location of the check-in.
- The system must be able to store and retrieve learner attendance records.
- The system must allow an admin user to create and manage cohorts. The system must allow an admin user to set location, start and end dates for each cohort. The system must be able to generate daily attendance reports for each cohort.
- The system must be able to notify learners of successful check-ins. The system must be secure and protect learner data.
- The system must be scalable to accommodate growth in the number of learners and data being processed.

2.5-2. Non-functional Requirements

- **Performance:** The system should be able to handle a high volume of check-in requests simultaneously without any lag or delay. The system should also be able to match the learner's face to the pre-uploaded image in real-time, within 2 seconds. The system should be optimized for quick retrieval of data, ensuring that the admin can access the attendance data anytime they need it.
- **Scalability:** The system should be able to handle many learners and images without degrading performance or causing downtime. The system should be able to accommodate growth in the number of learners and the amount of data being processed.
- **Security:** The system should implement strong encryption and authentication measures to protect the personal data of learners and to prevent unauthorized access to the system. The system should be regularly tested for vulnerabilities and any identified security issues should be promptly addressed with patches and updates.
- **Availability:** The system should be highly available and accessible, with minimal downtime and interruptions. The system should always provide accurate and up-to-date information on attendance.
- **Usability:** The system should be user-friendly and easy to navigate for both learners and admin users. The system should have clear instructions and provide helpful feedback to users.
- **Maintainability:** The system should be easy to maintain and update. The code should be well-documented and modular, allowing for easy modifications and enhancements. Compatibility: The system should be compatible with a wide range of devices and browsers, ensuring that learners can easily access the system from their preferred device. The system should also be able to integrate with other software tools used by the institution.
- **Reliability:** The system should be reliable and consistent in its performance, with minimal errors and downtime. The system should also be able to recover from failures or errors without significant loss of data or functionality.

2.6-Project Considerations

2.6-1. Project Constraints

Constraints: Design constraints consist of the Functionality of the solution. Project constraint consists of a functional model of presented design & the course constraints consists of lack of prior knowledge regarding the project's topic and technologies used for this solution.

2.6-2. Project Limitations

Using Agile Methodology

Using python for Facial recognition and AI development

Using ReactJs for web development alongside using Bootstrap.

2.6-3. Project Delimitations

Hardware limitations:

The system's performance may be limited by the hardware used to run it. For example, if the server hardware is not powerful enough to handle many users, it could affect the system's performance and availability.

Security limitations:

The use of facial recognition technology and the storage of personal information makes the system vulnerable to potential security threats. If the system is not designed and implemented with strong security measures, it could be subject to data breaches and other security issues.

User acceptance:

The success of the system depends on the acceptance and adoption by the users, including the learners, instructors, and administrators. If users find the system difficult to use or do not trust the system's accuracy, it could lead to low adoption rates and limited effectiveness.

Legal and ethical limitations:

The use of facial recognition technology and the collection and storage of personal data are subject to legal and ethical considerations. The system must comply with relevant laws and regulations and adhere to ethical standards for the use of personal data. Failure to do so could result in legal and reputational risks for the organization.

Integration limitations:

The system may need to integrate with other systems or platforms used by the organization, such as a learning management system or an identity management system. Integration may require additional resources and technical expertise and could affect the system's performance and availability.

2.6-4. Project Standards

[IEEE/ISO/IEC P15288 - ISO/IEC/EEE Draft Standard - Systems and Software engineering -- System Life Cycle Processes](#)

This International Standard establishes a common framework of process descriptions for describing the life cycle of systems created by humans. It defines a set of processes and associated terminology from an engineering viewpoint. These processes can be applied at any level in the hierarchy of a system's structure. Selected sets of these processes can be applied throughout the life cycle for managing and performing the stages of a system's life cycle. This is accomplished through the involvement of all stakeholders, with the goal of achieving customer satisfaction. This International Standard also provides processes that support the definition, control and improvement of the system life cycle processes used within an organization or a project. Organizations and projects can use these processes when acquiring and supplying systems. This International Standard concerns those systems that are made by humans and may be configured with one or more of the following system elements: hardware, software, data, humans, processes (e.g., processes for providing service to users), procedures (e.g., operator instructions), facilities, materials and naturally occurring entities. Ref. 3

CS423 chap 1

CS423 chap 2

2.7-Conclusion

In conclusion, this project involves the development of a web-based attendance management system using facial recognition technology and AI. The project constraints include functionality, functional model, and lack of prior knowledge, while the project limitations involve using Agile methodology, Python, and React.js for development. The project delimitations include hardware limitations, security concerns, user acceptance, legal and ethical considerations, and integration with other systems. To ensure high-quality standards and customer satisfaction, the project will adhere to the IEEE/ISO/IEC P15288 standards for systems and software engineering life cycle processes. It is essential to consider these limitations and constraints while developing and implementing the system to ensure successful adoption and effectiveness by the users.

3- Product Design

3.1- Introduction

This section will contain a description of the system's architecture starting from the high level, ending with detailed description on how the system should work as well as how modules interact with each other.

3.2- High-Level Design

3.2-1. Architecture Overview

In this project we are going to be using a three-tier architecture, Users or “learners” use the Client Dashboard (Web App) UI to interact with the application. Admin and staff use the Admin Dashboard UI.

The application is developed using python, React.JS, OpenCV.

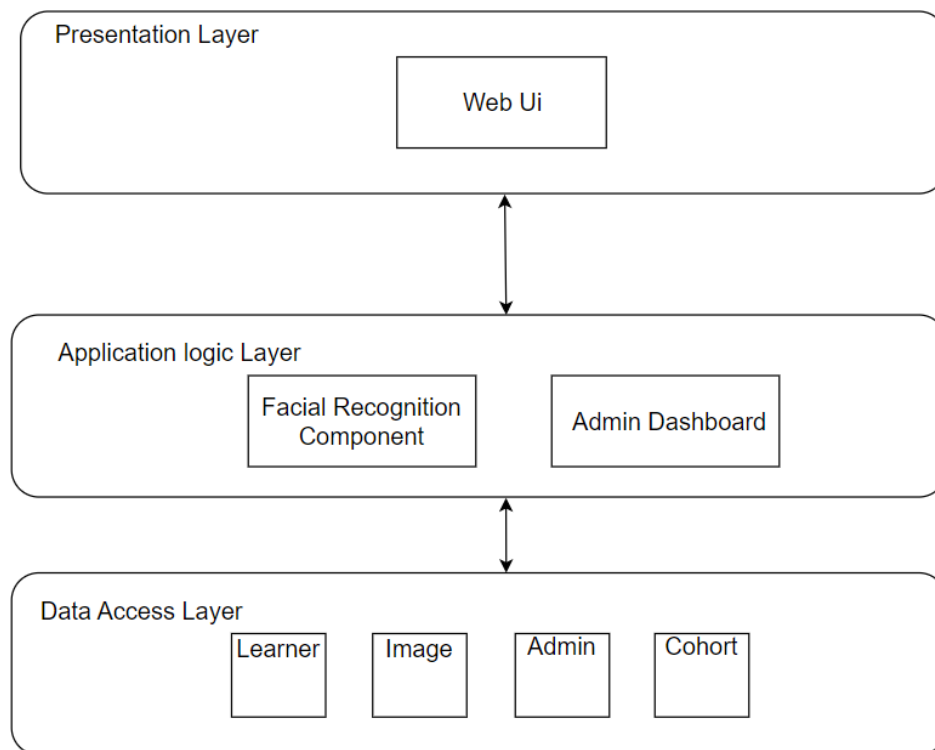


Figure 2: 3-tier architecture Diagram

3.2-2. Component Diagrams

The component diagram below shows the main modules of our project attendance tracking system and the interaction between its components.

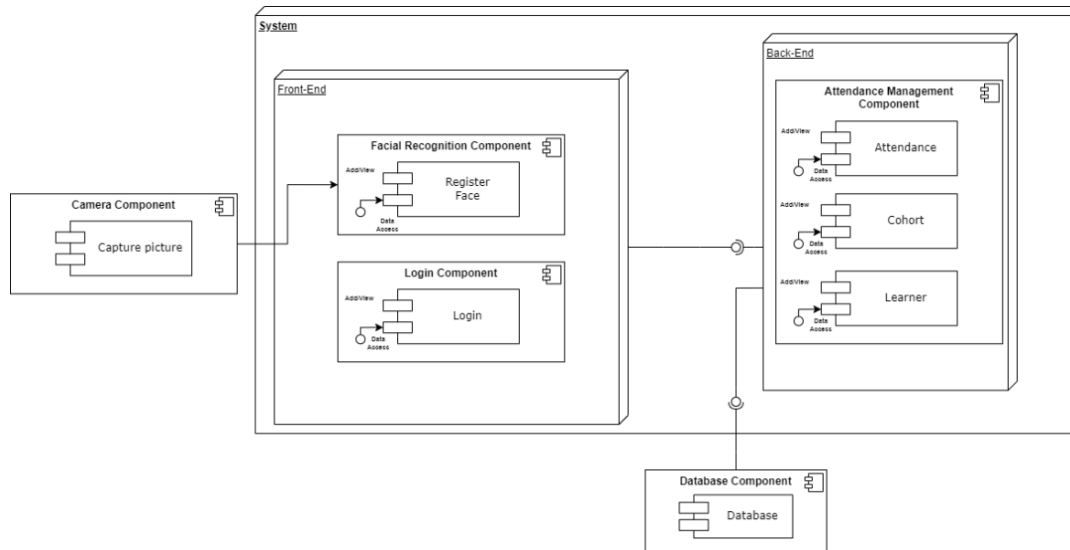


Figure 3 - Component Diagram

3.3- Detailed Design

The use case diagram below contains details about the use cases that are deemed to be architecturally significant for our system, other details are omitted for their simplicity and redundancy.

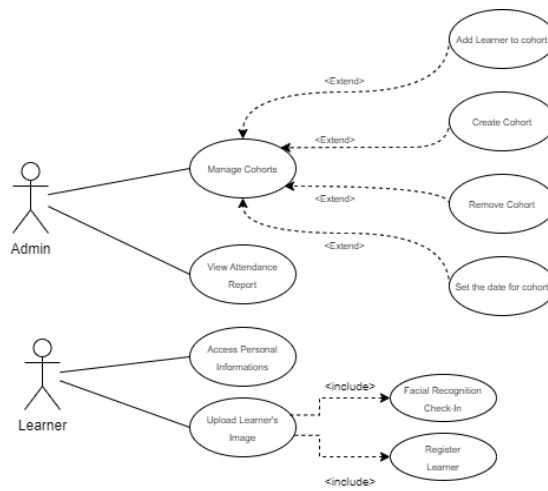


Figure 4 - Use Case Diagram

Use Case Scenarios:

UC0: Upload learner image:

- The learner wants to capture and upload an image to the system database to register.

Primary Actor:

- learner

Pre-condition:

- None

Post condition:

- The image is processed and saved to the database.
- Learner is registered.

Basic Flow:

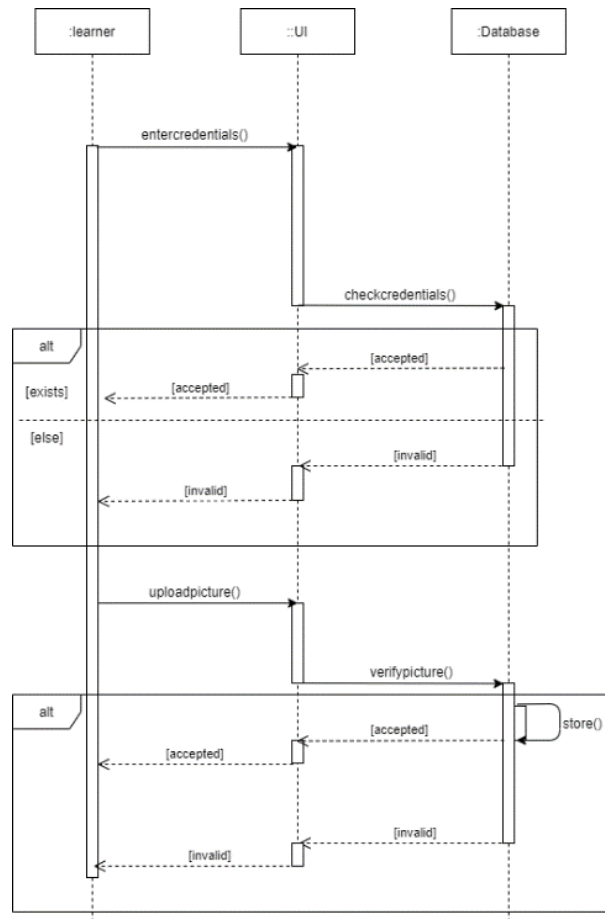
1. The use case begins when the learner visits the learner interface.
2. The learner clicks the “login” button.
3. The learner takes pictures for facial recognition.
4. The system validates the data entered.

Alternative Flow:

1. If the learner is unable to capture a clear image due to low lighting or other environmental factors, the system may prompt the learner to move to a better-lit location or adjust the camera settings to capture a clearer image.

Exceptions:

- The system does not validate the pictures.
- Accounts already exist, the user is asked to login with the facial recognition system.
- **SSD0: Upload learner image:**



UC1: Facial Recognition Check-In:

- The user wants to check in to the ODC to access cohort.

Primary Actor: Learner

Pre-condition:

- The learner has registered for the cohort and uploaded a photo of themselves.
- The learner is on site at the ODC. The system is online and operational.

Post condition:

- The learner's attendance record has been updated with the check-in information.
- The system has accurately recorded the learner's check-in and location.

Basic Flow:

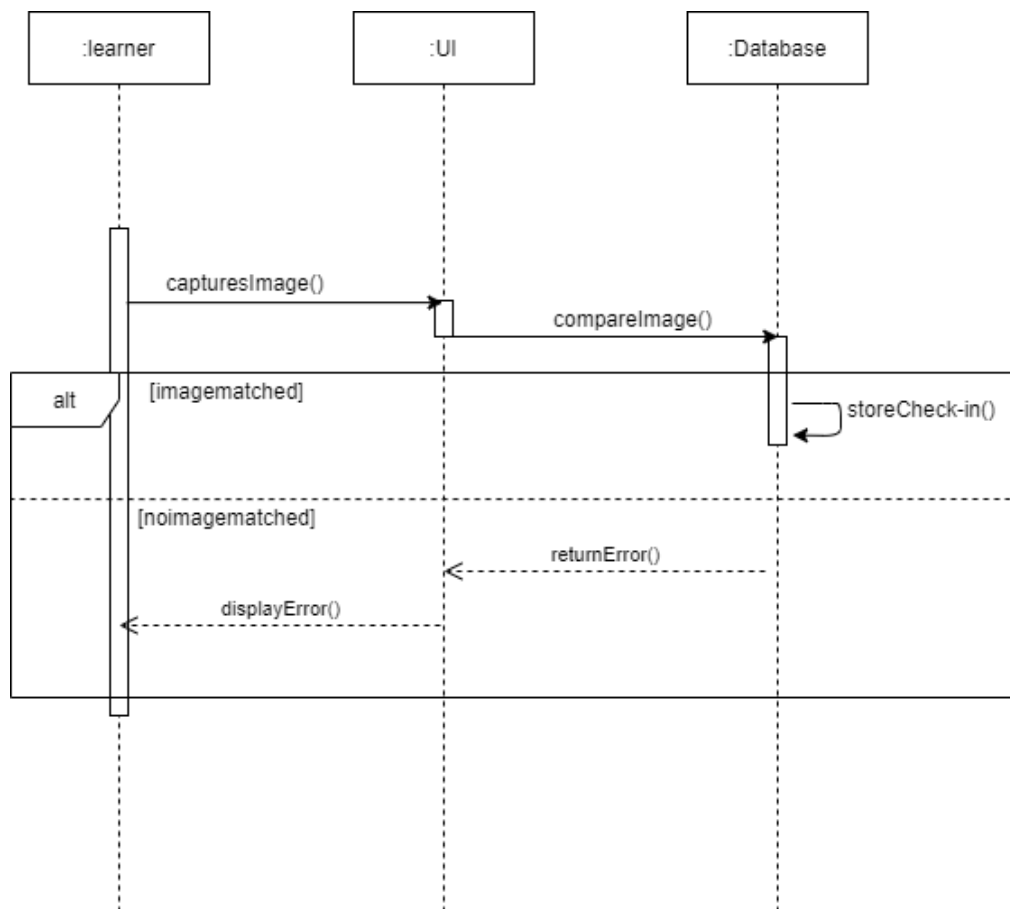
1. The learner approaches the check-in station and activates the check-in process.
2. The system captures an image of the learner's face using a webcam or similar device.
3. The system compares the captured image to the pre-uploaded image stored in the system.
4. If the images match, the system displays a message indicating a successful match and stores the check-in information, including the date, time, and location.

Alternative Flow:

1. If the images do not match, the system displays a message indicating an unsuccessful match and prompts the learner to try again or seek assistance.

Exceptions:

- If the system is unable to connect to the database or if there is a technical failure, the check-in process will be disrupted, and the learner will be unable to check in.
- If the system is unable to match the learner's face to the pre-uploaded image, the learner will need to try again or seek assistance.

SSD1: Facial Recognition Check-In:**UC2: View Personal Attendance Report:**

- The learner can view their attendance report for the current cohort, which displays the dates and times they have checked in.

Primary Actor: Learner

Pre-condition:

- The learner has already been enrolled in a cohort and has completed the check-in process at least once.
- The attendance records for the cohort have been updated in the system.

Post condition:

- The learner has successfully viewed their attendance report and confirmed their attendance information.

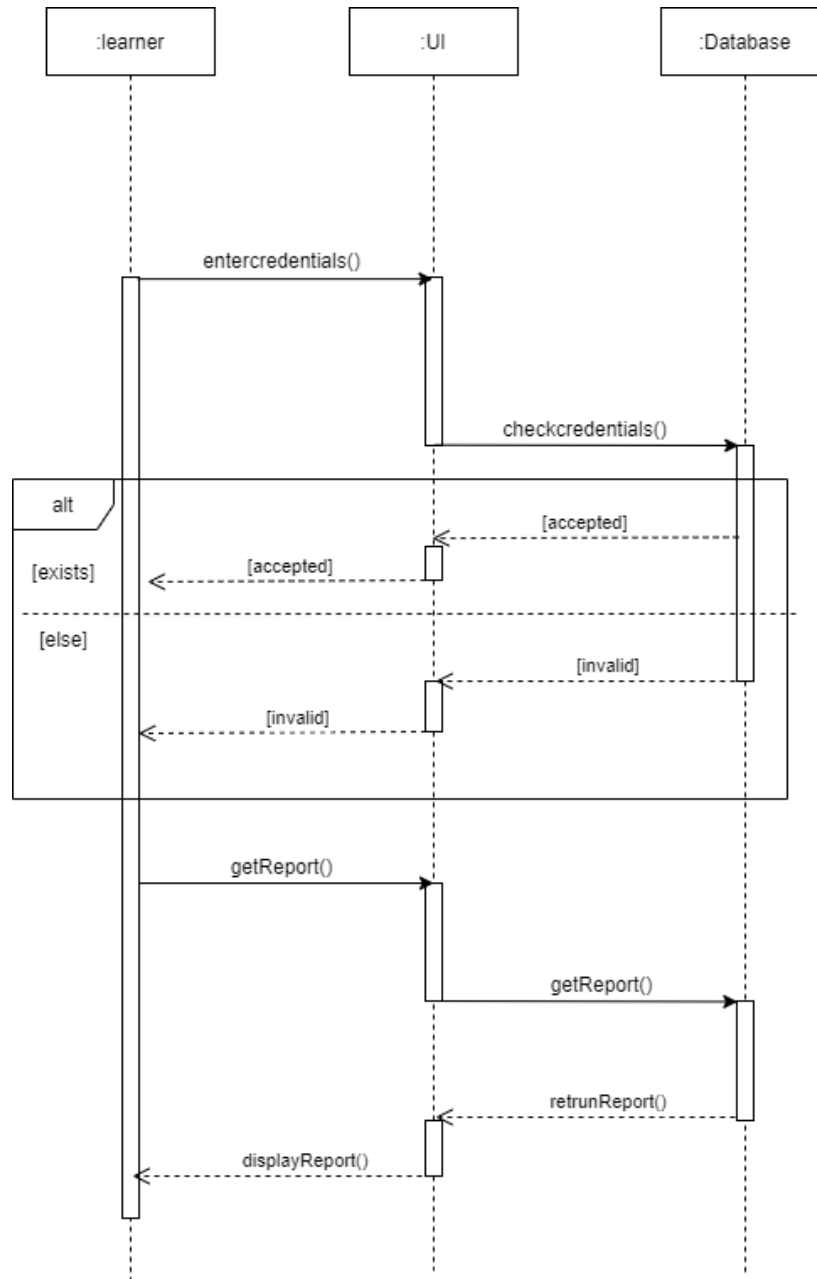
Basic Flow:

1. The learner logs into the system using their credentials.
2. The learner navigates to the "Attendance Report" page.
3. The system displays the learner's attendance report, which includes the dates and times they have checked in.
4. The learner can view the report and confirm their attendance information. The learner logs out of the system.

Exceptions:

- Learner will receive an error message indicating that the system is unavailable and to try again later.
- Learner will receive an error message indicating that their attendance data is missing or incorrect.
- Learner will receive an error message indicating what is wrong with the information they provided.

- **SSD2: View Personal Attendance Report:**



UC3: Manage Cohort:

- The admin user can create a new cohort in the system, specifying the location, start and end dates, and other relevant details, the admin can also add or remove a specific user to the cohort, update the time of the said cohort.

Primary Actor: Admin

Pre-condition:

- The admin user is logged into the system.
- The admin user has sufficient permission to manage and edit cohorts.

Post condition:

- The new cohort is created and is available for use in the system.
- The details for the new cohort are stored in the system for future reference.
- The cohort can be updated.

Basic Flow:

1. The admin user accesses the "Create New Cohort" feature in the system.
2. The admin user enters the details for the new cohort, including the location, start and end dates, and any other relevant information.
3. The admin user clicks the "Save" button to create the new cohort.
4. The system confirms the creation of the new cohort and displays the details for the new cohort.

Sub Flow:

1. After 4 the admin can access the “edit cohort” feature which allows him to change the time chosen for the cohort
2. After 4 the admin can access the “edit cohort” feature which allows him to add a new user to the cohort
3. After 4 the admin can access the “edit cohort” feature which allows him to remove a user from the cohort.
4. After 4 the admin can access the “edit cohort” feature which allows him to delete the cohort.

3.4- UI Design

In this portion we are going to be showcasing the UI of our project

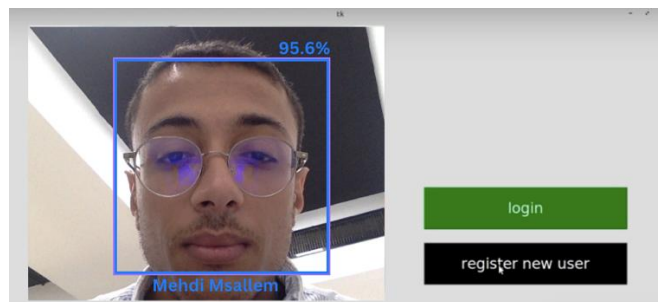


Figure 6 -- UI Interface Facial Recognition

3.5- Conclusion

4- Development and Prototype Solution

4.1- Introduction

In this following chapter, we will be providing an overview of the implementation specifics of the Attendance tracker. The information provided will encompass comprehensive particulars about the various technologies that are being utilized for the development of the front-end, backend, and database of the project.

4.2- Development Details

4.2.1- Frontend technologies

To develop a web application that is both user-friendly and efficient, we opted to use the ReactJs framework for our front-end development. This decision was based on several factors, including the need for a scalable and fast solution that could handle the complexities of a large web application. Moreover, ReactJs offers a component-based approach that aligns with our project requirements, making it an ideal choice for our development needs.

4.2.2- Backend technologies

For the development of the application's backend, we decided to use NodeJS as our runtime environment and ExpressJS as our framework. We made this choice because we found these technologies to be the most appropriate for our project's specific requirements. Additionally, the extensive documentation available for NodeJS and ExpressJS was extremely helpful in guiding us through the development process. We used MySQL as the primary data store for our application since it allows us to store the data in a structured format.

4.2.3- Database system

We opted for MySQL which is an open-source relational database management system. MySQL's versatility and scalability make it an excellent option for storing large and complex data securely. MySQL's reliability and performance have made it one of the most widely used database

management systems for web applications, providing us with a dependable and robust data storage solution.

4.3- Development Support

To ensure that our application performs at its best, we have put in place a range of support mechanisms such as maintenance, updates, and bug fixes. These measures are implemented to ensure that our application continues to function efficiently and effectively over time. Our goal is to provide a seamless user experience by addressing any issues that may arise and ensuring that our application remains up to date with the latest developments in technology.

-Maintenance: Our plan to maintain the performance, security, and scalability of our application is to conduct system checks, optimize performance, and maintain the database on a regular basis. By regularly reviewing the system's health, we can take proactive measures to ensure that any potential issues are addressed promptly, thereby minimizing downtime and maximizing the application's performance.

- Updates: We are going to be improving the application by releasing frequent updates that will include fixes and more enhancements based on our system checks and the users' feedback.

- Bug fixes: As a committed team we will proactively track and resolve any reported glitches or problems within the application. Our process involves a methodical approach to detect, examine, and rectify bugs promptly to reduce any interruptions in the application's performance.

-Future development: We are going to enhance the application by adding new features and transforming it into a learning management system.

4.4- Conclusion

To conclude, this chapter presents a general summary of the attendance tracker's creation and prototype solution. It describes the technologies utilized for the app's frontend, backend, and database. The chapter also emphasizes the support structures in effect, such as maintenance, updates, and bug fixes, to guarantee the uninterrupted functionality of the application. Furthermore, it acknowledges continuous efforts to enhance the app's performance and lays out strategies for future development and improvements.

5- Tests, Results, and discussions

5.1 Introduction

The following chapter details the tests executed on the attendance tracker application, the corresponding findings, and ensuing discussions. It thoroughly evaluates the application's usability, functionality, and performance through meticulous testing and analysis.

5.2 Discussion

5.2.1- Testing Methodology

The testing phase for the attendance tracker comprised multiple techniques, such as smoke testing and usability testing. web testing aimed to ensure the application meets the functional As well as the non-functional requirements. The goal is to identify bugs and defects in the application and to ensure that it is working as expected.

5.2.2- Testing Results

Here are some of the results of the smoke testing conducted:

test cases:

1-testing login

test environment:

operating system: windows 10

browser version: Microsoft edge version

test data: valid username and password

test execution: manual testing

test results

all expected results

test status: Pass

test environment:

operating system: windows 10

browser version: Microsoft edge version

test data: invalid username and password

test execution: manual testing

test results:

all expected results "invalid login"

test status: pass

2-testing facial recognition

test environment:

operating system: windows 10

browser version: Microsoft edge version

test data: no face displayed in facial recognition

test execution: manual testing

test results

all expected results "face front camera"

test status: pass

test environment:

operating system: windows 10

browser version: Microsoft edge version

test data: valid facial recognition

test execution: manual testing

test results

all expected results

test status: pass

test environment:

operating system: windows 10

browser version: Microsoft edge version

test data: invalid facial recognition

test execution: manual testing

test results

all expected 'face not recognized'

test status: pass

5.3 Conclusion

In conclusion, the testing phase for our attendance tracking management system with facial recognition utilized web testing, mostly manual test to verify the validity of the functional and

nonfunctional requirements, regression testing is planned for future work, after implementing more features.

6- General Conclusion & Future work

In conclusion, the implementation of the Attendance Tracking Management System with Facial Recognition has successfully addressed the issues faced by the Orange Development Center in managing attendance. The web-based platform has reduced the time and effort required to track attendance, minimized errors and inconsistencies, and provided valuable insights into attendance patterns. Facial recognition technology has eliminated the need for manual check-ins, increasing efficiency and accuracy. The project has also provided a user-friendly interface for both learners and admin users, making it easy to manage attendance and cohorts.

Future Work:

Although the project has been successfully implemented, there is always room for improvement. In the future, the system can be further developed by adding more features such as automatic email notifications for absent learners, real-time alerts for tardiness. Moreover, the system's security can be enhanced by integrating multi-factor authentication, such as a password and facial recognition or a fingerprint scan for the admin dashboard. Additionally, the system can be integrated with other existing platforms, such as learning management systems, to provide a more comprehensive solution for Orange Development Center. Overall, the Attendance Tracking Management System with Facial Recognition has provided a solid foundation for further development and enhancement.

References
